

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EE)

November 2013

EE208 Micro Electronics & Integrated Circuit

Date: 28.11.2013, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Write down definition of a Transistor. Draw symbol of Transistor and describe each pin of Transistor. [07]

- Q - 2 (a) Draw and explain characteristics of common emitter connection of Transistor. [07]
(b) Explain biasing of a Transistor with Base Resistor Method. [07]

OR

Q - 2 (a) What do you mean by h parameters? Find out h parameters in terms of voltage and current. [07]

(b) Explain working principle of JFET with neat diagram. [07]

Q - 3 (a) What is the meaning of Faithfull Amplification of a Transistor? What are the conditions must be satisfied for that? Explain any two. [07]

(b) Draw the h parameter equivalent circuit for Transistor. Find out Input impedance, Current gain and Voltage gain in terms of h parameters for Transistor. [07]

OR

Q - 3 (a) Write down differences between JFET and Bipolar Transistor. [05]

(b) Explain following terms with necessary diagram. [06]

- (i) Shorted-gate drain current
- (ii) Pinch off Voltage
- (iii) Gate-source cut off voltage

(c) Write down applications of JFET. [03]

SECTION - II

Q - 4 Write down the electrical characteristics of an ideal op-amp. [07]

Q - 5 (a) Draw and explain current-to-voltage converter using op-amp. [04]

(b) Explain Averaging Amplifier in inverting configuration with neat circuit diagram. [07]

(c) Draw and explain Subtractor. [03]

OR

Q - 5 (a) Draw the frequency response curve of following Active filters. [06]

(i) Low-pass filter

(ii) High-pass filter

(iii) Band-pass filter

(iv) Band-reject filter

(b) Explain 555 timer as an Astable Multivibrator. [08]

Q - 6 (a) Draw and explain equivalent circuit for an op-amp. Also draw ideal voltage transfer-curve. [07]

(b) Find out closed-loop voltage gain (A_F) equation for inverting Amplifier with neat circuit diagram. [07]

OR

Q - 6 (a) Draw and explain all-pass filter. [07]

(b) Draw pin diagram and Internal Structure block diagram of 555 timer IC. [07]
